

Engineering Project Report

Project 078: STM32 ARM Cortex-M4 Real-Time FFT Audio Spectrum Analyzer

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Abstract

This technical report presents the comprehensive design, circuit schematics, bill of materials, mathematical modeling, assembly methodology, and experimental testing for **STM32 ARM Cortex-M4 Real-Time FFT Audio Spectrum Analyzer**. Classified under the **Advanced / Expert (Engineering Level)** tier in the **DSP / Embedded Systems** category, this design provides optimized performance, high reliability, and clear practical utility for school and engineering applications.

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1 Introduction & System Overview

The objective of this project is to implement a robust electronic system for **STM32 ARM Cortex-M4 Real-Time FFT Audio Spectrum Analyzer**. This document details the step-by-step engineering design, component functions, mathematical principles, and experimental results.

2 Bill of Materials (Components List)

Table 1: Bill of Materials for STM32 ARM Cortex-M4 Real-Time FFT Audio Spectrum Analyzer

S.No.	Component Name	Qty	Circuit Function
1	Power Transistors / High-End MCU	2	Power MOSFETs / IGBTs / STM32 / ESP32
2	High-Frequency Transformers	1	Ferro-magnetic energy storage core
3	Gate Drivers / Fast Diodes	2	IR2110 gate driver and Ultrafast diodes
4	Sensing Amplifiers / CT Sensors	2	Current sensors (ACS712/SCT-013) and Differential amps
5	Filter Capacitors (Low ESR)	2	Low ESR electrolytic and Polypropylene film caps
6	Auxiliary Power Protection	1 Set	Auxiliary SMPS, heatsinks, and snubber networks

3 System Architecture & Methodology

The STM32 ARM Cortex-M4 Real-Time FFT Audio Spectrum Analyzer is an advanced engineering implementation featuring high-frequency switching, closed-loop feedback algorithms, or high-speed DSP processing for maximum efficiency and precision.

4 Circuit Assembly & Testing Procedure

1. Place core active components and microcontrollers/ICs on the breadboard or PCB.
2. Wire DC supply rails (+VCC and GND) with appropriate decoupling capacitors.
3. Connect sensor networks and input signal conditioning circuits.
4. Interface output switches, MOSFET drivers, or visual/acoustic indicators.
5. Apply power and measure operating node voltages to verify proper performance.

5 Mathematical Derivations & Governing Equations

$$V_{\text{out}} = \frac{V_{\text{in}}}{1 - D} \quad (1)$$

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\% \quad (2)$$

$$f_r = \frac{1}{2\pi\sqrt{LC}} \quad (3)$$

6 Applications & Engineering Conclusion

Industrial automation, renewable energy systems, electric vehicle power trains, aerospace electronics, and high-performance dsp / embedded systems.